

1. Class D addresses are designated for multicast. Explain why a typical corporation requiring addresses for its individual workstations and servers cannot use a block of Class D addresses for this purpose. (4 marks)
2. What is the fundamental difference between a *subnet mask* and a *default mask* in IPv4 addressing? Provide an example of each for a Class B network. (4 marks)
3. An ISP has allocated the address block 192.168.40.0/22 to a small business. Which of the following host addresses would be routable within this business's network? Justify your answer. (4 marks)
 - I. 192.168.39.255
 - II. 192.168.41.10
 - III. 192.168.43.1
 - IV. 192.168.40.254
4. A university department is granted the IP block 172.16.32.0/20. They need to create at least 12 subnets, with the largest subnet requiring 150 hosts. (4 marks)
 - I. What is the new subnet mask that allows for this?
 - II. How many usable host addresses will each of these new subnets have?
 - III. What is the network address and broadcast address of the 3rd usable subnet?
5. The TCP/IP model is often compared to the OSI model. In a single, clearly labeled diagram, map the four layers of the TCP/IP model to the seven layers of the OSI model, showing their correspondence. (4 marks)

1. Why Class D Addresses Cannot Be Used for Workstations/Servers

Class D addresses (224.0.0.0-239.255.255.255) are *reserved exclusively for multicast traffic*. They do not represent unicast host addresses and therefore:

- Cannot be assigned to individual devices such as workstations or servers.
- Do not support network and host portions in the same way unicast addresses do.
- Routers treat them as multicast group identifiers, **not routable unicast endpoints**.

Conclusion:

A corporation cannot assign Class D addresses to host devices because they are not valid unicast addresses.

2. Difference Between Subnet Mask and Default Mask (Class B Example)

Default Mask

- The mask *inherent to the class* of the IP address.
- For Class B: 255.255.0.0 (/16)

Subnet Mask

- A customized mask used to divide the network into subnets.
- Extends beyond the default mask.

Example (Class B subnet mask):

- 255.255.240.0 (/20)

Fundamental difference:

- Default mask identifies only the *network class boundary*.
- Subnet mask breaks the address further into subnet and host portions.

3. Routable Addresses for 192.168.40.0/22

Step 1: Determine block size

/22 → block size = 4 in the third octet ($256 - 252 = 4$).

Network range:

- Network: 192.168.40.0
- Broadcast: 192.168.43.255
- Usable range: 192.168.40.1 → 192.168.43.254

Evaluate each address:

Address	Routable?	Reason
I. 192.168.39.255	☒ No	Outside /22 (below 192.168.40.0)
II. 192.168.41.10	☒ Yes	Inside usable range
III. 192.168.43.1	☒ Yes	Inside usable range
IV. 192.168.40.254	☒ Yes	Inside usable range

Correct answers: II, III, and IV

4. Subnetting 172.16.32.0/20

Step I — New Subnet Mask

Requirements:

- At least 12 subnets
- Largest subnet needs 150 hosts (needs ≥ 8 host bits $\rightarrow 2^8 - 2 = 254$ usable)

Start with /20 and borrow bits:

- /20 $\rightarrow$ /24 gives:
 - 16 subnets (2^4) $\rightarrow$ meets 12 requirement
 - 254 hosts $\rightarrow$ meets 150 requirement

✓ New subnet mask = /24 (255.255.255.0)

Step II — Usable Host Addresses Per Subnet

/24 $\rightarrow$ 8 host bits

Usable = $2^8 - 2 = 254$

Step III — 3rd Usable Subnet Addresses

/24 subnets of the original /20 start at intervals of 1 in the fourth octet:

- Subnet 1: 172.16.32.0/24
- Subnet 2: 172.16.33.0/24
- Subnet 3: 172.16.34.0/24

Network address: 172.16.34.0

Broadcast address: 172.16.34.255

5. TCP/IP Model vs OSI Model Diagram

+-----+-----+		
TCP/IP Model	OSI Model	
+-----+-----+		
Application Layer	Application	
	Presentation	
	Session	
+-----+-----+		
Transport Layer	Transport	
+-----+-----+		
Internet Layer	Network	
+-----+-----+		
Network Access Layer	Data Link	
	Physical	
+-----+-----+		

To determine which of the listed IPs are valid hosts within this block, we must first understand the size and range of a /22 network.

Step 1 — Determine the Network Range of 192.168.40.0/22

A /22 mask means:

- Subnet mask: 255.255.252.0
- The block size in the third octet is:
 $256 - 252 = 4$
- Therefore, subnets increase in steps of 4 in the third octet.

The block containing 192.168.40.0 starts at 40 because:

$40 \div 4 = 10$ (an exact boundary)

So the network spans:

- Network address: 192.168.40.0
- Next network: 192.168.44.0
- Therefore, the broadcast address is 1 less:

Broadcast address: 192.168.43.255

Usable host range:

192.168.40.1 → 192.168.43.254

Any address inside this range is a valid routable host.

Step 2 — Evaluate Each Address

I. 192.168.39.255

☒ NOT routable

Reason:

- This address is below the network start (192.168.40.0).
 - It belongs to the previous /22 block (192.168.36.0/22).
-

II. 192.168.41.10

☒ Routable

Reason:

- Falls between 192.168.40.1 and 192.168.43.254.
-

III. 192.168.43.1

☒ Routable

Reason:

- Still within the valid range.
 - It is not the broadcast address (43.255).
-

IV. 192.168.40.254

☒ Routable

Reason:

- Within the first subnet segment of the block.

- Not a network or broadcast address.

Final Answer

Address	Routable?	Explanation
I. 192.168.39.255	☒ No	Outside 192.168.40.0/22 range
II. 192.168.41.10	☒ Yes	Inside valid host range
III. 192.168.43.1	☒ Yes	Inside valid host range
IV. 192.168.40.254	☒ Yes	Inside valid host range

✓ Routable addresses: II, III, IV